MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics | **CBSE CLASS 10 – PERIODIC TEST 08** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 9: Some Applications of Trigonometry** | **MM: 30**

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, correct algebraic substitution, and final boxed answer with proper physical units.
2. In alternative (**OR**) questions, step marking applies equally to both options.
3. Partial marks are allocated per step as indicated in brackets: **[0.5M, 1M, 2M]**. Alternative mathematically valid methods are fully credited.

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Multiple Choice & Assertion-Reason

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Q1. The line drawn from the eye of an observer to the point in the object viewed is termed as: [1]

(A) Horizontal line (B) Line of sight (C) Normal ray (D) Vertical plumb line

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B)
Justification: By standard geometric definition, the ray from the observer's eye to the viewed point on the object is called the Line of sight.

Q2. An observer looks up at an aeroplane flying in the sky. The angle between the horizontal line through the eye and the line of sight is called: [1]

(A) Angle of depression (B) Angle of elevation (C) Vertical angle (D) Reflected angle

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (B)
Justification: When the viewed object lies above the horizontal eye-level line, the acute angle formed is the Angle of elevation.

Q3. An observer standing on top of a cliff looks down at a boat in the sea. The angle between the horizontal and line of sight is: [1]

(A) Angle of depression (B) Angle of elevation (C) Acute horizon (D) Zenith angle

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]
Correct Option: (A)
Justification: When the viewed object lies below the horizontal eye-level line, the acute angle between horizontal and line of sight is the Angle of depression.

Q4. If a vertical pole of height 18 m casts a shadow on the ground such that $\tan\theta = \frac{6}{7}$, then the length of the shadow is: [1]

(A) 21 m (B) 15.4 m (C) 24 m (D) 18 m

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: $\tan\theta = \frac{18}{\text{shadow}} = \frac{6}{7} \implies \text{shadow} = \frac{18 \times 7}{6} = \boxed{21 \text{ m}}$.

Q5. **Assertion (A):** If a ladder leaning against a wall makes an angle of 30° with the ground, then the distance of its foot from the wall is greater than the height reached on the wall.
Reason (R): For $\theta = 30°$, $\cot 30° = \sqrt{3} \approx 1.732 > 1$, meaning base $= \sqrt{3} \times$ height $>$ height. [1]

(A) Both (A) and (R) are true; (R) explains (A).
(B) Both are true; (R) does not explain (A).
(C) (A) is true but (R) is false.
(D) (A) is false but (R) is true.

► **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: (A)

Justification: Since base $= h\cot 30° = \sqrt{3}\,h > h$, Assertion (A) is true and Reason (R) gives the exact mathematical proof. Both (A) and (R) are true and (R) explains (A).

SECTION B: Very Short Answer Questions

MATHS MANTHAN | ($2 \times 2 = 4$ Marks)

Q6. A tightrope walker climbs a rope of length 20 m attached from the ground to the top of a pole. If the rope makes an angle of 30° with the ground level, find the height of the pole. [2]

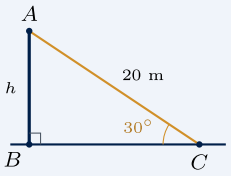
► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Let pole $AB = h$ m and rope $AC = 20$ m inclined at $\angle ACB = 30°$. [0.5 Mark]

In right $\triangle ABC$: $\sin 30° = \frac{AB}{AC} \implies \frac{1}{2} = \frac{h}{20}$. [1 Mark]

$$\implies h = \boxed{10 \text{ m}}$$

Final pole height: [0.5 Mark]



Q7. The angle of elevation of the top of a tower from a point on the ground is 30°. If the observer moves 20 m towards the tower, the angle of elevation becomes 60°. Find the height of the tower. **[2]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 2 Marks]

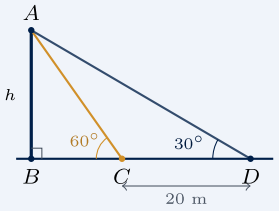
Let tower $AB = h$ m, initial point D (30°), closer point C (60°) with $CD = 20$ m and $BC = x$ m. [0.5 Mark]

In $\triangle ABC$: $x = \dfrac{h}{\sqrt{3}}$; in $\triangle ABD$: $x + 20 = h\sqrt{3}$. Subtracting:

$20 = h\left(\sqrt{3} - \dfrac{1}{\sqrt{3}}\right) = \dfrac{2h}{\sqrt{3}}$. [1 Mark]

$\implies h = 10\sqrt{3} = 10(1.732) = \boxed{10\sqrt{3} \text{ m} \approx 17.32 \text{ m}}$

Final tower height: [0.5 Mark]



SECTION C: Short Answer Questions

MATHS MANTHAN | $(2 \times 3 = 6$ Marks)

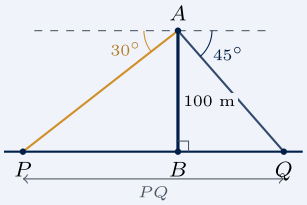
Q8. As observed from the top of a 100 m high lighthouse from the sea level, the angles of depression of two ships on opposite sides of the lighthouse are 30° and 45°. If the line joining the ships passes through the base of the lighthouse, find the distance between the two ships. [Take $\sqrt{3} = 1.732$] **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let lighthouse $AB = 100$ m, with opposite ships P (depression 30° $\implies \angle APB = 30°$) and Q (depression 45° $\implies \angle AQB = 45°$). [0.5 Mark]

In $\triangle ABP$: $PB = 100 \cot 30° = 100\sqrt{3}$ m. [1 Mark]

In $\triangle ABQ$: $BQ = 100 \cot 45° = 100$ m. [1 Mark]



Distance between the two ships $= PQ = 100\sqrt{3} + 100 = 100(\sqrt{3} + 1) = 100(2.732) = \boxed{273.2 \text{ m}}$.

[0.5 Mark]

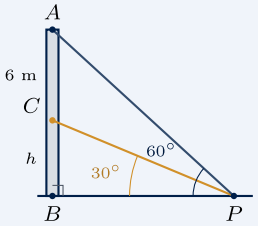
Q9. A vertical flagstaff 6 m high stands on top of a pedestal. From a point on the ground, the angles of elevation of the bottom and top of the flagstaff are 30° and 60° respectively. Find the height of the pedestal. **[3]**

► STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 3 Marks]

Let pedestal be $BC = h$ m and flagstaff be $AC = 6$ m, viewed from point P with $\angle CPB = 30°$ and $\angle APB = 60°$. [0.5 Mark]

In right $\triangle CBP$: $\tan 30° = \dfrac{h}{BP} \implies BP = h\sqrt{3}$ — (i). In right $\triangle ABP$: $\tan 60° = \dfrac{h+6}{BP} \implies BP = \dfrac{h+6}{\sqrt{3}}$ — (ii).

[1 Mark]

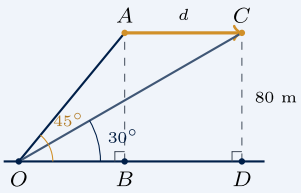


Equating (i) and (ii): $h\sqrt{3} = \dfrac{h+6}{\sqrt{3}} \implies 3h = h + 6 \implies 2h = 6 \implies \boxed{h = 3 \text{ m}}$. [1.5 Marks]

Q10. A bird is sitting on the top of an 80 m high tree. From a point on the ground, the angle of elevation of the bird is 45°. The bird flies away horizontally at a constant height. After 2 seconds, the angle of elevation of the bird from the same observation point is 30°. Find the flying speed of the bird in m/s and km/h. [Take $\sqrt{3} = 1.732$] **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let tree $AB = 80$ m, observer at O, initial bird position A ($\angle AOB = 45°$), position after 2 s be C above D at constant height $CD = 80$ m ($\angle COD = 30°$). [1 Mark]



In right $\triangle ABO$: $\tan 45° = \dfrac{80}{OB} \implies OB = 80$ m. In right $\triangle CDO$: $\tan 30° = \dfrac{80}{OD} \implies OD = 80\sqrt{3}$ m. [1.5 Marks]

Distance flown in 2 s: $AC = BD = OD - OB = 80\sqrt{3} - 80 = 80(1.732 - 1) = 80 \times 0.732 = 58.56$ m. [1 Mark]

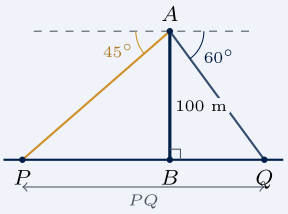
$$\text{Speed (m/s)} = \frac{58.56}{2} = \boxed{29.28 \text{ m/s}}, \qquad \text{Speed (km/h)} = 29.28 \times \frac{18}{5} = \boxed{105.41 \text{ km/h}}$$

Evaluating both speed units: [1.5 Marks]

Q11. From the top of a tower 100 m high, an observer sees two cars on opposite sides of the tower on a straight horizontal road passing through the base. If their angles of depression are 45° and 60°, find the distance between the two cars. [Take $\sqrt{3} = 1.732$] **[5]**

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

Let tower $AB = 100$ m, and opposite cars be at P (depression $45° \implies \angle APB = 45°$) and Q (depression $60° \implies \angle AQB = 60°$). [1 Mark]



In right $\triangle ABP$: $\tan 45° = \dfrac{100}{PB} \implies PB = 100$ m. [1 Mark]

In right $\triangle ABQ$: $\tan 60° = \dfrac{100}{BQ} \implies BQ = \dfrac{100}{\sqrt{3}} = \dfrac{100\sqrt{3}}{3}$ m. [1 Mark]

Distance between the two cars $PQ = PB + BQ = 100 + \dfrac{100\sqrt{3}}{3} = 100 + \dfrac{173.2}{3} = 100 + 57.733$ m.

$$\implies PQ = 100\left(1 + \frac{\sqrt{3}}{3}\right) \approx \boxed{157.73 \text{ m}}$$

Rationalizing and summing distances: [2 Marks]

Q12. CASE STUDY: Hot Air Balloon Festival Optical Tracking

A tethered research balloon A ascends vertically at altitude h directly above launch pad B. Two optical tracking stations P (nearer, elevation $60°$) and Q (farther, elevation $30°$) are stationed 100 m apart along the same radial line ($PQ = 100$ m).

(i) Express the horizontal ground distance from station P to the launch pad (BP) in terms of the altitude h. [1]

(ii) Express the horizontal ground distance from station Q to the launch pad (BQ) in terms of the altitude h. [2]

(iii) Calculate the exact vertical altitude h of the balloon. [Take $\sqrt{3} = 1.732$] [2]

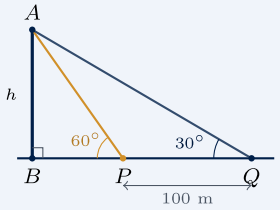
OR

Find the numerical horizontal distance of tracking station P from the launch pad B. [2]

▶ STEP-BY-STEP SOLUTION & MARKING SCHEME: [Total: 5 Marks]

(i) In right $\triangle ABP$ ($\angle APB = 60°$):

$$\tan 60° = \frac{h}{BP} \implies BP = \boxed{\frac{h}{\sqrt{3}}}$$



[1 Mark]

(ii) In right $\triangle ABQ$ ($\angle AQB = 30°$):

$$\tan 30° = \frac{h}{BQ} \implies BQ = \boxed{h\sqrt{3}}$$

Subpart (ii): [2 Marks]

(iii) Main Choice: Since $PQ = BQ - BP = 100$ m, $h\sqrt{3} - \dfrac{h}{\sqrt{3}} = 100 \implies \dfrac{2h}{\sqrt{3}} = 100 \implies h = 50\sqrt{3} = 50(1.732) = \boxed{50\sqrt{3} \text{ m} \approx 86.60 \text{ m}}$. [2 Marks]

(iii) OR Choice: Substituting $h = 50\sqrt{3}$ m into $BP = \dfrac{h}{\sqrt{3}}$ gives $BP = \dfrac{50\sqrt{3}}{\sqrt{3}} = \boxed{50 \text{ m}}$. [2 Marks]

★★★ END OF SOLUTIONS · MATHS MANTHAN MEERUT ★★★